8.1.1 Actions Taken Based on the Results of Evaluation of the COs Attainment (20)

1. Areas of Weakness Identified:

a. Attainment for slow learners: Certain students consistently underperformed, lowering overall course attainment averages.
b. Difficulty in problem-based courses: Students found it hard to integrate knowledge from multiple subjects when solving complex problems.
c. Communication and teamwork-related COs attainment: Group projects and presentations revealed gaps in technical report writing and effective communication.

2. Measures Identified and Implemented for Improvement

Curriculum Interventions

a. Introduced mini-projects to enhance application and design-oriented learning.
b. Updated laboratory manuals and experiments to include simulation-based and industrial problems.
c. Integrated skill-based modules (software tools, industry case studies, etc.) into selected courses.

Pedagogical Initiatives

a. Adoption of active learning techniques (quiz-based learning, peer learning).
b. Increased focus on problem-solving sessions and assignments linked with industrial problems.
c. Regular use of ICT tools (NPTEL/SWAYAM resources, simulation software, online interactive platforms) to supplement teaching.

Support System Improvements

a. Establishment of bridge/remedial classes for slow learners.
b. Mentoring and tutorial hours to support weaker students.
c. Strengthened interactions with outside world through guest lectures.
d. e-resource access improved to support self-learning.

Monitoring and Feedback

a. Continuous monitoring of CO attainment through periodic tests.
b. Collection of student feedback and alumni/industry feedback to identify skill gaps.
c. Implementation of corrective actions in subsequent semesters (course content adjustments, more practice problems).

8.1.2 Actions Taken Based on the Results of Evaluation of the POs/PSOs Attainment (20)

The following criterion was set to assess the performance of each course in terms of PO attainment:

< 50%	Poor
50% - 60%	Average
60% - 70%	Good
70% - 80%	Very Good
> 80%	Excellent

POs & PSOs Attainment Levels and Actions for improvement based on 2023 batch:

POs	Target Level	Attainment Level	Observations
PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.			
PO1	75.00	85.4	**Observations:** **1.** Overall attainment is above the target. **2.** In only 03 courses, attainment was less than 75%. **3.** In courses, namely CAD, Thermodynamics, and Automobile lab, attainment was between 65-75 %.
Action to improve attainment: Thermodynamics and CAD are core mechanical engineering courses and thus require more effort to achieve an attainment of more than 80%. For this, it is planned to provide more tutorials to the upcoming batches. Weak students shall be advised to consult the respective subject teachers to clarify their doubts. Based on the day-to-day performance of the Automobile Lab, the students scoring below 70% attainment, it was planned that the weak students would undergo rigorous experimental practice for further improvement.			
PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.			

PO2	75.00	85.46	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, Thermodynamics, and Automobile lab, attainment was between 65-75 %.
Action to improve attainment: Thermodynamics and CAD are core mechanical engineering courses and thus require more effort to achieve an attainment of more than 80%. For this, it is planned to provide more tutorials to the upcoming batches. Weak students shall be advised to consult the respective subject teachers to clarify their doubts. Based on the performance of the Automobile Lab, the students scoring below 70%, more emphasis will be given on results and analysis to improve their attainment.			
PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.			
PO3	75.00	85.42	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, Thermodynamics, and Automobile lab, attainment was between 65-75 %.
Action to improve attainment: Thermodynamics and CAD are core mechanical engineering courses and thus require more effort to achieve an attainment of more than 80%. For this, it is planned to provide more tutorials to the upcoming batches. Weak students shall be advised to consult the respective subject teachers to clarify their doubts. Based on the day-to-day performance of the Automobile Lab, the students scoring below 70% attainment, it was planned that the weak students would undergo rigorous experimental practice for further improvement.			
PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods, including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.			

PO4	75.00	85.3	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, Thermodynamics, Power Plant Engineering, and Automobile lab, attainment was between 65-75 %.
Action to improve attainment: Thermodynamics, Power Plant Engineering, and CAD are core mechanical engineering courses and thus require more effort to achieve an attainment of more than 80%. For this, it is planned to provide more tutorials to the upcoming batches and encourage students peer learning through discussions. Weak students shall be advised to consult the respective subject teachers to clarify their doubts. Based on the performance of the Automobile Lab, the students scoring below 70%, more emphasis will be given for extended lab classes for hands-on practice.			
PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities with an understanding of the limitations.			
PO5	75.00	85.43	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, Thermodynamics, and Automobile lab, attainment was between 65-75 %.
Action to improve attainment: Thermodynamics and CAD are core mechanical engineering courses and thus require more effort to achieve an attainment of more than 80%. Extra time will be devoted to weak students to clarify their doubts. Link the experiments with industrial case studies to improve motivation. Encourage peer mentoring and frequent oral questioning during lab to strengthen conceptual clarity.			
PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.			

PO6	75.00	85	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, and Thermodynamics attainment was between 65-75 %.
Action to improve attainment: To be improved through remedial sessions, enhanced practical training, and industry-oriented exposure.			
PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for, sustainable development.			
PO7	75.00	84.8	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, and Thermodynamics attainment was between 65-75 %.
Action to improve attainment: Assignments and projects related to environmental sustainability.			
PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.			
PO8	75.00	84.16	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, Thermodynamics, and Automobile lab, attainment was between 65-75 %.
Action to improve attainment: 1. Conduct mandatory lab safety and ethics orientation at the beginning of the course. 2. Organize interactive sessions/seminars on professional ethics.			

PO9: Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.			
PO9	75.00	84.84	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, Thermodynamics, and Automobile lab, attainment was between 65-75 %.
Action to improve attainment: Promote an interactive classroom session. Motivate students to participate in individual events such as paper presentations in technical seminars. Introduce group projects to cultivate collaborative learning.			
PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as comprehending and writing effective reports and design documentation, making effective presentations, and giving and receiving clear instructions.			
PO10	75.00	84.94	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, Thermodynamics, and Automobile lab, attainment was between 65-75 %.
Action to improve attainment: 1. Teacher to encourage students to participate in the question-answer session/teacher-student interaction. 2. Encourage students to publish their project works in thermal engineering in the various conferences and scientific communities.			
PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.			

PO11	75.00	84.59	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, and Thermodynamics, attainment was between 65-75 %.
Action to improve attainment: The institute provides financial support to students through the department, depending on their project/internship requirements.			
PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.			
PO12	75.00	85.36	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD and Thermodynamics, attainment was between 65-75 %.
Action to improve attainment: 1. Encourage self-learning – PowerPoint presentation, Project report on Frontier areas of mechanical engineering. 2. Advise students on e-learning resources on technological changes in various areas of mechanical engineering.			
PSO1: The student will be able to design mechanical systems in various fields and challenges such as machine elements, thermal systems, manufacturing, and industrial and inter disciplinary fields like additive manufacturing, soft computing to meet the demand of the day in industry as well as society.			

PSO1	75.00	85.47	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, Thermodynamics, and the Automobile Lab, attainment was between 65-75 %.
Action to improve attainment: 1. Strengthening analysis in the laboratory courses. 2. Guest Lectures on Scale-up in the design of the mechanical industry. 3. Take up more design-based projects in the field of Mechanical Engineering.			
PSO2: The student will be capable to start their entrepreneurship, be employable and suitable for various fields like design, manufacturing, production industries, academic and industries, research, and development organizations in and outside the country.			
PSO2	75.00	85.52	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD, Thermodynamics, and the Automobile Lab, attainment was between 65-75 %.
Action to improve attainment: 1. Strengthening analysis in the laboratory courses. 2. Students will be encouraged to take up more Mechanical Engineering-related projects (targeting manufacturing and design areas).			
PSO3: The student will be able to pursue advanced degrees in engineering, business, or other professional fields through their knowledge and learned skills through formal as well as informal self-study and motivation.			

PSO3	75.00	87.45	**Observations:** 1. Overall attainment is above the target. 2. In only 03 courses, attainment was less than 75%. 3. In courses, namely CAD and Thermodynamics, attainment was between 65-75 %.
Action to improve attainment: 1. Students are to be encouraged to take up collaborative projects with students and faculty of other Departments. 2. Students will be encouraged to take up Industry collaborative projects with joint guidance of BIT faculty and Industry for the final semester credited project.			

In general, the following actions are planned to improve the attainment level:

1. Assignments/Tutorials related to real-life problems and industrial problems.
2. Surprise quiz on important topics.
3. Special focus on weak students through extra classes, tutorials, and assignments.
4. Guest lecture by Industry personnel.

8.2 Academic Audit and actions taken thereof during the period of Assessment (15)

The institution has established a structured **Academic Audit system** to ensure systematic evaluation and enhancement of academic processes.

The process involves:

Collection of Data: In a standardised format from Departments.

Assessment and Score Calculation: As per Academic Audit Guidelines

SWOT Analysis and Recommendations: First at Departmental Level as well as at Institute Level.

External committee evaluation: Finally the evaluation and action plan is to be done by the external committee duly constituted by the Institute.

Name of Campus						
Name of Department:						
				Placement and Higher studies		
Student Strength (If more than one program either in UG or PG, please input the summation value)	**AY 2024-25**			**Graduate Students**	**AY 2024-25**	
	Sanctioned Strength	**Enrolled Students Strength**		Students Enrolled in Final Year (UG+PG)		
UG Program (Year 1)				Students Graduated (UG+PG)		
UG Program (Year 2)				How many of these students were placed?		
UG Program (Year 3)				How many of these opted for higher studies?		
UG Program (Year 4)						
UG Program (Year 5) if applicable				Total No. of PhD students in the Department		
Total UG	0	0		No. of PhD students graduated		
PG Program (Year 1)						
PG Program (Year 2)						
Total PG	0	0		**Outreach Activities**	**Y/N**	**Count (in AY 2024-25)**
Total (UG + PG)	0	0		No. of Conferences/Workshops//Seminars etc. organized by the Department		
				No. of faculty members who visited abroad		
Faculty Qualification	**AY 2024-25**			No. of faculty members who attended Conferences, Workshops, FDPs, MDPs, Short Term Training Programs etc. within India		
Total Faculty in 2023-24				No. of MoUs (all active MoUs with Activity Reports)		
How many faculty with PhD degree?				No. of Awards/Recognitions received by faculty		
How many faculty with Post graduate degree?				No. of faculty who delivered Invited Talks		
Financial Resource Utilization	**FY 2024-25**					
				Accreditation Related Activities (Part A)	**Count (AY 2024-25)**	
Amount of Budget allocated to Department (Rs)				Number of Departmental Societies and Club Activities		
Amount Utilized (Rs)				Number of Professors of Practice and Chair Professors		
				Number of New courses introduced		
Research and Professional Practices	**Calendar Year 2024**			Number of books and book chapters published by the department		
No. of Unique Publications published by the Department						
No. of publications in WoS database?						
No. of publications only in Scopus database but not in WoS database?				**Accreditation Related Activities (Part B)**	**Y/N**	
No. of Patents Granted (exclude design patent)				Implementation of Feedback Analysis and Action Taken Report		
No. of Patents Published (exclude design patent)				Implementation of Mentor-Mentee System		
				Implementation of OBE in all programs (applicable to programs eligible for NBA)		
Amount (Rs) received from	**FY 2024-25**					
Govt. Sponsored R&D Projects						
Consultancy Projects						
CSR Projects						
External agencies or registration for participation to organize FDPs, MDPS, EDPs etc.						
Govt. agencies or external agencies for Organizing Certificate Programmes / Training Workshops/Seminars/confernce/symposium etc.		**Date**	**Place**	**Name and Signature of Head of Department**		
Total	0					

Guidelines for Academic Audit:

Summary of Academic Audit Parameters and Weightages

Sr. No.	Parameter	Marks	Weightage	Net marks
1	Teaching, Learning & Resources (TLR)	100	0.30	30
2	Research and Professional Practice (RP)	100	0.30	30
3	Graduation Outcomes (GO)	100	0.20	20
4	Outreach Activities (OA)	100	0.10	10
5	Accreditation Related Activities (ARA)	100	0.10	10
Total Academic Audit Score				100

S.No.	Parameters	Marks
1.	**Teaching, Learning & Resources (TLR) Weight: 0.30**	**100**
	Student Strength (SS): 20 marks Faculty Student Ratio (FSR): 30 marks Faculty Qualification Index (FQI): 20 marks Financial Resource Utilization (FRU): 30 marks	
2.	**Research and Professional Practice (RP)** **Weight: 0.30**	**100**
	Publications (PU): 35 marks Quality of Publications (QP): 40 marks IPR and Patents: Published and Granted (IPR): 15 marks Footprint of Projects and Professional Practice (FPPP): 10 marks	
3.	**Graduation Outcomes (GO)** **Weight: 0.20**	**100**
	Placement and Higher Studies (GPH): 50 marks University Examinations (GUE): 25 marks Number of Ph.D. Students Enrolled and Graduated (GPHD): 25 marks	
4.	**Outreach Activities (OA)** **Weight: 0.10**	**100**
	As per various Outreach Activities undertaken by Department	

5.	**Accreditation Related Activities (ARA)** **Weight: 0.10**	**100**
As per various Accreditation Related Activities undertaken by Department		

1.Teaching, Learning & Resources (TLR): 100 marks

Weight: 0.30

Overall Assessment Metric:

TLR = SS (20) + FSR (30) + FQI (20) + FRU (30)

Component metrics based on:

Student Strength: SS

Faculty-Student Ratio: FSR

Faculty with PhD: FQI

Financial Resource Utilization: FRU

Student Strength (SS): 20 Marks

SS = [No. of students enrolled / Sanctioned Intake] x 100

SS	**Marks**
>= 90%	20
80% to 89.99%	17
70% to 79.99%	14
60% to 69.99%	11
50% to 59.99%	08
40% to 49.99%	05
Less than 40%	00

Data to be collected:

	Previous Academic Year (N)	**Enrolled student strength (E)**
Sanctioned Strength of UG Program (Year 1)		
Sanctioned Strength of UG Program (Year 2)		

Sanctioned Strength of UG Program (Year 3)		
Sanctioned Strength of UG Program (Year 4)		
Sanctioned Strength of UG Program (Year 5) If applicable		
Total UG (X)		
Sanctioned Strength of PG Program (Year 1)		
Sanctioned Strength of PG Program (Year 2)		
Total PG (Y)		
Total (X+Y)	N=	E=
SS	(E/N)*100 =	

A. **Faculty-Student Ratio (FSR): 30 marks**

•FSR = N/F

- ***N = No. of students (UG + PG) in previous Academic Year***
- F: Faculty (who have taught at least 2 semesters) in the institution in the previous academic year. *Faculty on contract basis/ad-hoc basis will be considered if the concerned faculty has taught in both the semesters of previous Academic Year*

- FSR

FSR	Marks
Less than 15	30
15.01 to 16.99	26
17.00 to 18.99	22
19.00 to 20.99	18
21.00 to 22.99	14
23.00 to 25.00	10
Greater than 25.00	00

Data to be collected:

	Previous
Sanctioned Strength of UG Program (Year 1)	
Sanctioned Strength of UG Program (Year 2)	
Sanctioned Strength of UG Program (Year 3)	
Sanctioned Strength of UG Program (Year 4)	
Sanctioned Strength of UG Program (Year 5) If applicable	

Total UG (X)	
Sanctioned Strength of PG Program (Year 1)	
Sanctioned Strength of PG Program (Year 2)	
Total PG (Y)	
Total (X+Y)	N=
F (No. of Faculty) *to be used in all metrics requiring Total No. of Faculty*	
FSR (N/F)	

B. **Faculty Qualification Index (FQI): 20 marks**

Faculty qualification index (FQI) = 2 * [(10X +4Y)/F)]

 X=No. of faculty members with Ph.D. degree or equivalent as per AICTE/UGC norms.

 Y=No. of faculty members with M. Tech. or ME degree or equivalent as per AICTE/UGC norms.

 F=Total No. of faculty

Data to be collected:

	Previous Academic Year
No. of faculty with PhD (X)	
No. of faculty with Post Graduate degree (Y)	

D: Financial Resources and their Utilisation (FRU): 30 Marks

- **FRU = [Amount of allocated budget utilized / Budget allocated] x 100**

FRU(%)	**Marks**
>= 90%	30
80% to 89.99%	27
70% to 79.99%	24
60% to 69.99%	21
50% to 59.99%	18
40% to 49.99%	15
Less than 40%	00

Data to be collected:

	Previous Financial Year
Amount of Budget allocated to Department (B)	
Amount utilized U)	
FRU{(U/B)*100}	

2. Research and Professional Practice (RP): 100 marks

- **Weight: 0.30**
- **Overall Assessment Metric:**

For Architecture, Management, HSS, CQEDS, Maths, RS:

RP = PU (40) + QP (40) + FPPP (20)

For Other Departments:

RP = PU (35) + QP (40) + IPR (15) + FPPP (10)

- **The component metrics:**

A. **Total Publications: PU**
B. **Quality of Publications: QP**
C. **IPR and Patents: Patents Published and Granted: IPR**
D. **Footprint of Projects and Professional Practice: FPPP**

A. **Publications (PU): 35 marks / 40 marks (for Arch, HSS, CQEDS, Mgmt, Maths, RS)**
   - **PU = Total Unique Publications made by the department (P) / No. of Faculty (F)**

If PU is:

For other Departments	For Architecture, Management, HSS, CQEDS, Mathematics, Remote Sensing
>= 3.0 – 35 marks	>= 3.0 – 40 marks
2.5 – 3.0 – 30 marks	2.5 – 3.0 – 35 marks
2.0 – 2.5 – 25 marks	2.0 – 2.5 – 30 marks
1.5 – 2.0 - 20 marks	1.5 – 2.0 - 25 marks
1.0 – 1.5 – 15 marks	1.0 – 1.5 – 20 marks
0.5 – 1.0 – 10 marks	0.5 – 1.0 – 15 marks
Less than 0.5 – 0 marks	Less than 0.5 – 0 marks

Data to be collected:

	Previous calendar Year
Total unique Publications made by the department (P)	
PU = (P/F)	

B. **Combined metric for Quality of Publications (QP): 40 Marks (for all)**

QP = [Publications in SCI/SCIE (R) / Total unique Publications (P)] x 100

100% - 40 marks

80% - 90% - 38 marks

70% - 80% - 36 marks

60% - 70% - 32 marks

50% - 60% - 30 marks

40% - 50% - 28 marks

30% - 40% - 26 marks

20% to 30% - 24 marks

10% to 20% - 22 marks

less than 10% - 20 marks

Data to be collected:

	Previous calendar Year
Total unique Publications made by the Department (P)	
How many of these publications were in WoS database? (R)	
QP = (R/P)*100	

C. IPR and Patents: Patents Published and Granted (IPR): 15 marks

(metric not applicable to Architecture, Management, HSS, CQEDS, Maths, RS)

- **IPR = [2 x Granted Patents (G) + Published Patents (P)] / No. of Faculty (F)**

IPR ≥ 0.5 – 15 marks

0.4 ≤ IPR < 0.5 – 12 marks

0.3 ≤ IPR < 0.4 – 9 marks

0.2 ≤ IPR < 0.3 – 6 marks

0.1 ≤ IPR < 0.2 – 3 marks

IPR < 0.1 – 0 marks

Data to be collected:

	Previous Calendar Year
No. of Patents Granted (G)	
No. of Patents Published (P)	
IPR = (2G+P)/F	

D. Footprint of Projects and Professional Practice (FPPP): 10 marks / 20 marks (for Arch, HSS, CQEDS, Management and Mathematics)

•FPPP = (SPR + C + CSR + FDP + WS) / F

If FPPP is:

For other Departments	For Architecture, Management, HSS, CQEDS, Mathematics, Remote Sensing
> 5 lakhs – 10 marks	> 5 lakhs – 20 marks
4 to 5 lakhs – 8 marks	4 to 5 lakhs –18 marks
3 to 4 lakhs – 6 marks	3 to 4 lakhs – 16 marks
2 to 3 lakhs – 4 marks	2 to 3 lakhs – 14 marks
1 to 2 lakhs – 2 marks	1 to 2 lakhs – 12 marks
Less than 1 lakh – 0 marks	Less than 1 lakh – 0 marks

Data to be collected:

Amount received from	Amount received in the previous Financial Year
Govt. Sponsored R&D Projects (SPR)	
Consultancy Projects (C)	
CSR Projects (CSR)	
External agencies or registration for participation to organize FDPs, MDPS, EDPs etc. (FDP)	
Govt. agencies or external agencies for Organizing Certificate Programmes / Training Workshops/Seminars/confernce/symposium etc. (WS)	
Total Amount (X)	
FPPP = (X/F)	

3. **Graduation Outcome (GO):100 marks**

a. **Weight: 0.20**
b. **Overall Assessment Metric:**

c.GO = GPH (50) + GUE (25) + GPHD (25)

The component metrics are explained on the following pages:

A. **Placement and Higher Studies: GPH**
B. **Metric for University Examinations: GUE**
C. **Metric for Number of Ph.D. Students Enrolled and Graduated: GPHD**
D. **Metric for Placement and Higher Studies (GPH): 50 marks**
   - **GPH = 50 x [(Np / 100)+(Nhs / 100)]**

Np = (No. of Students Placed / No. of Students Graduated) x 100

Nhs = (No. of Students opted for Higher Studies / No. of Students Graduated) x 100

Data to be collected:

	Previous Academic Year
No. of Graduate Students (*to be used in next metric also*)	
No. of Students Placed	
No. of Students opted for Higher Studies	

B. Metric for University Examinations (GUE): 15 Marks

- **GUE = 25 × min [(N_g/80), 1]**
- Ng = (No. of students graduated / No. of students enrolled in final year) x 100

Data to be collected:

	Previous Academic Year
No. of Students enrolled in Final Year	

C. Metric for Number of Ph.D Students Enrolled and Graduated (GPHD): 25 Marks

GPHD = (N1/F) + (N2/F) [N1 = No. of PhD Students Enrolled; N2 = No. of PhD Students Graduated

If (N1/F) is:

>3 – 10 marks

2 to 3 – 8 marks

1 to 2 – 6 marks

Between 0 to 1 – 4 marks

No scholar – 0 marks

If (N2/F) is:

>1.0 – 15 marks

0.8 to 1.0 – 8 marks

0.6 to 0.8 – 6 marks

0.4 to 0.6 – 4 marks

0.2 to 0.4 – 3 marks

< 0.2 – 0 marks

Data to be collected:

	Previous Academic Year
No. of PhD Students Enrolled (N1)	
No. of PhD Students Graduated (N2)	

4. Outreach Activities (OA): 100 marks

Weight: 0.10

Overall Assessment Metric has the following components:

a. Conferences / Workshops / Seminars etc. organized by the Department
b. Faculty members visited abroad
c. Faculty members attended Conferences, Workshops, FDPs, MDPs, Short Term Training Programs etc. within India
d. MoUs (all active MoUs with Activity Reports)
e. Awards/Recognitions received by faculty
f. Faculty delivered Invited Talks

Number of Outreach activities	Marks
All 6	100
Only 5	80
Only 4	60
Only 3	40
Only 2	20
Less than 2	00

Data to be collected:

Outreach activities
Number of Conferences / Workshops / Seminars etc. organized by the department
Number of Faculty members visited abroad
Number of Faculty members attended Conferences, Workshops, FDPs, MDPs, Short Term Training Programs etc. within India
Number of MoUs (all active MoUs with Activity Reports)
Number of Awards/Recognitions received by faculty
Number of Faculty delivered Invited Talks

4. **Accreditation Related Activities (ARA): 100 marks**

Weight: 0.10

Overall Assessment Metric has the following components:

a. Departmental Societies and Club Activities
b. Professors of Practice and Chair Professors
c. New courses introduced
d. Implementation of Feedback Analysis and Action Taken Report
e. Implementation of Mentor-Mentee System
f. Implementation of OBE in all programs (applicable to programs eligible for NBA)
g. Books and book chapters published by the department

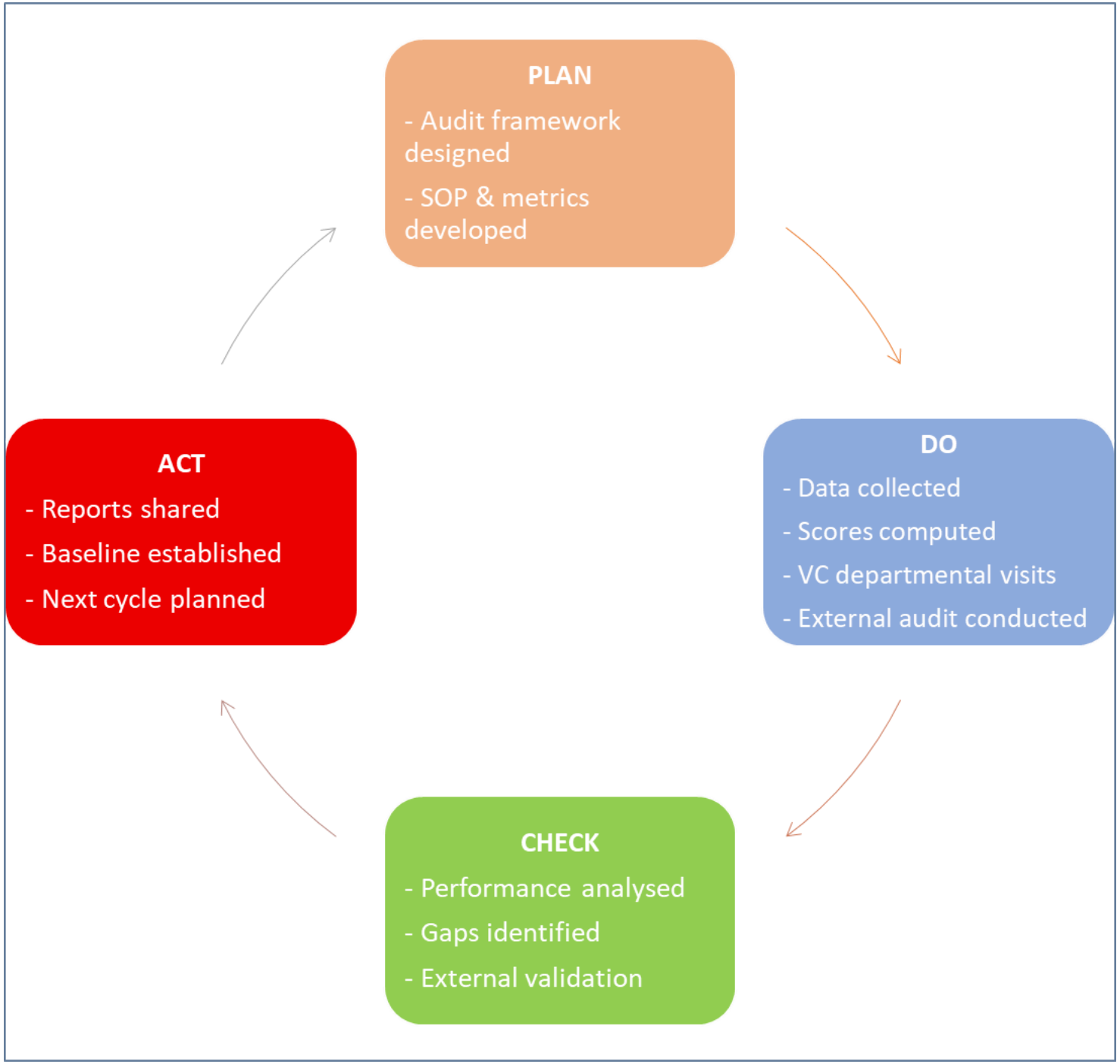
Number of Outreach activities	Marks
All 7	100
Only 6	80
Only 5	60
Only 4	40
Only 3	20

2 or less	00

Data to be collected:

Accreditation Related Activities (Part A)	**Count (A.Y. 2023-24)**
Number of Departmental Societies and Club Activities	
Number of Professors of Practice and Chair Professors	
Number of New courses introduced	
Number of books and book chapters published by the department	
Accreditation Related Activities (Part B)	**Y/N**
Implementation of Feedback Analysis and Action Taken Report	
Implementation of Mentor-Mentee System	
Implementation of OBE in all programs (applicable to programs eligible for NBA)	

PDCA Cycle for Academic Audit Initiative (IQAC)



Institution: Birla Institute of Technology, Mesra

Initiative: Academic Audit Framework Implementation

1. PLAN (P)

Objective:

To establish a systematic and quantitative framework for assessing academic performance across departments and ensuring quality enhancement.

Planning Activities:

1. IQAC developed a structured Academic Audit framework with defined parameters and weightages:
    i. Teaching, Learning & Resources (TLR)
    ii. Research & Professional Practice (RP)
    iii. Graduation Outcomes (GO)
    iv. Outreach Activities (OA)
    v. Accreditation Related Activities (ARA)
2. A detailed SOP and rubrics were formulated for uniform evaluation.
3. Standardised data collection templates (Excel-based) were designed for:
    i. Departmental data entry
    ii. Automated score calculation
4. Clear guidelines and timelines were communicated to all departments.

Expected Outcome:

1. Uniform and transparent academic performance assessment
2. Identification of strengths and areas of improvement
3. Support for accreditation processes (NAAC/NBA)

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2. DO (D)

Implementation:

1. Departments submitted data in prescribed formats.
2. Academic Audit Scores were computed based on defined metrics.

Administrative Review:

Vice Chancellor visited departments and reviewed:

i. Submitted data
ii. Academic practices
iii. Departmental performance

External Review:

An External Academic Audit Committee was constituted.

External experts:

i. Examined departmental data
ii. Interacted with HoDs and faculty
iii. Submitted audit reports and observations

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3. CHECK (C)

Evaluation:

1. Departmental performance analysed across all parameters
2. Comparative assessment carried out to identify:

i. Performance variations
ii. Key strengths and gaps

Validation:

Findings validated through:

i. VC observations
ii. External expert reports

Outcome of Check Phase:

Identification of:

i. Variability in research output and quality
ii. Differences in graduation outcomes
iii. Gaps in documentation and outreach activities

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4. ACT (A)

Action Taken:

1. Audit reports and observations were shared with all departments
2. Departments were informed of:
    i. Their performance scores
    ii. Observations by internal and external reviewers

Action Status

This cycle primarily focused on:

i. Establishing the audit system
ii. Conducting baseline assessment

Future Action Plan:

1. Departments are expected to:
    i. Analyse feedback
    ii. Initiate corrective measures
2. IQAC proposes to:
    i. Conduct subsequent audit cycles
    ii. Track improvements over time

Outcome of PDCA Cycle

- Successful implementation of a structured Academic Audit system
- Creation of a baseline for continuous quality improvement
- Enhanced transparency and accountability

8.3 Improvement in Faculty Qualification/Contribution (15)

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)	CAYm3 (2022-2
No. of faculty members with Ph.D. degree	25.00	23.00	22.00
No. of publications in peer reviewed journals	49.00	61.00	40.00
No. of publications in conferences	9.00	26.00	15.00

8.4 Improvement in Academic Performance (10)

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)
Academic Performance Index (API) of the First-Year Students in the Program (Refer to section 4.3)	6.73	5.93
Academic Performance Index (API) of the Second-Year Students in the Program (Refer to section 4.4)	8.54	7.55
Academic Performance Index (API) of the Third-Year Students in the Program (Refer to section 4.5)	8.72	8.59

9 STUDENT SUPPORT AND GOVERNANCE (120)

9.1 First Year Student-Faculty Ratio (FYSFR) (5)